EduPulse: Student Retention

& Dropout Analytics

Strategic Business Analysis Report

Target Audience: Executive Management, Client Stakeholders

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A. Executive Summary

Student attrition presents a significant operational and financial challenge for higher education institutions. The EduPulse project provides a data-driven framework to identify primary academic and financial drivers of dropout.

Key Validated Findings:
- Financial Criticality: Overdue tuition fees correlate with an 86.5% dropout rate.
- Academic Warning Signs: Dropout students average a 2nd-semester grade of 5.90 vs 12.70 for graduates.
- High-Risk Nexus: Combined poor performance (Grade < 8) and overdue fees lead to a 97% attrition rate.
- Protective Factors: Scholarship holders show only a 12.2% dropout rate.

Strategic Recommendation: Implement a Dual-Track Intervention System focusing on immediate financial outreach and 2nd-semester academic recovery programs.

B. Business Problem and Objectives

Student dropout is a critical business risk. High attrition leads to lost tuition revenue and decreased institutional prestige. EduPulse aims to identify the earliest reliable indicators of attrition to enable proactive intervention.

C. Data Overview

The analysis is based on 4,424 student records. Distribution of outcomes:
- Graduates: 49.93%
- Dropouts: 32.12%
- Enrolled: 17.95%

D. Key Business Findings

1. Financial Impact

Financial stress is the strongest leading indicator. Tuition fees overdue: 86.5% dropout; Debtors: 62% dropout. Scholarship holders: 12.2% dropout.

2. The High-Risk Segment

Students combining poor academic performance (Grade < 8) and overdue fees exhibit a 97% dropout rate, representing a near-certainty of attrition.

E. Academic Performance

A statistically significant gap exists in 2nd-semester performance. Graduates averaged 12.70 grades and 6.18 approved units, whereas dropouts averaged 5.90 grades and 1.94 approved units.

F. Financial and Demographic Patterns

Financial aid acts as a powerful stabilizer. The contrast between scholarship holders (12.2%) and non-holders (38.7%) suggests that removing financial anxiety directly impacts persistence.

G. Power BI Dashboard

The interactive dashboard enables management to explore these patterns:
- Executive Overview: Health check of student body by age and financial status.
- Dropout Drivers: Tracking the 2nd-semester 'danger zone'.
- Statistical Insights: Validated evidence ($p < 0.05$) to justify intervention budgets.

H. Recommendations and Action Plan

1. Critical: Tuition Alert System to prevent attrition from immediate financial crises.
2. High: 2nd-Semester Academic Bridge for students showing early performance dips.
3. Medium: Expansion of Scholarship programs to stabilize at-risk populations.

I. Reporting Automation

The Python-n8n-AI pipeline automates the flow from raw data to stakeholder email. This reduces reporting latency and translates complex statistics into plain-language business insights.

J. Limitations and Responsible Interpretation

Analysis identifies associations, not causation. Group patterns do not dictate individual fate. Recommendations should be balanced with institutional context.

K. Conclusion

EduPulse proves that retention is a manageable problem. By addressing financial instability and academic decline in the second semester, the institution can secure revenue and improve student success.